

CS6313- Mini Project 3

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Section 1

1. Suppose the lifetime, in years, of an electronic component can be modeled by a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

where $\theta > 0$ is an unknown parameter. Let $X_1, \dots, X_n$ be a random sample of size n from this population.

(a) Derive an expression for maximum likelihood estimator of θ .

a) Derive the MLE of θ
for a random sample $X_1, X_2, \dots, X_n$

$$L(\theta) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \frac{\theta}{x_i^{\theta+1}} = \theta^n \prod_{i=1}^n x_i^{-(\theta+1)}$$

$$\ell(\theta) = \ln L(\theta) = n \ln \theta - (\theta+1) \sum_{i=1}^n \ln x_i$$

$$\frac{d\ell}{d\theta} = \frac{n}{\theta} - \sum_{i=1}^n \ln x_i = 0$$

$$\frac{n}{\theta} = \sum_{i=1}^n \ln x_i \Rightarrow \theta = \frac{n}{\sum_{i=1}^n \ln x_i}$$

b) Suppose $n = 5$ and the sample values are $x_1 = 21.72$, $x_2 = 14.65$, $x_3 = 50.42$, $x_4 = 28.78$, $x_5 = 11.23$. Use the expression in (a) to provide the maximum likelihood estimate for θ based on these data.

$$\begin{aligned}
 & \text{b) } \sum_{i=1}^n \ln x_i \quad x_1 = 21.72, x_2 = 14.65, x_3 = 50.42 \\
 & \quad \quad \quad x_4 = 28.78, x_5 = 11.23 \\
 & \sum_{i=1}^n \ln x_i = \ln(21.72) + \ln(14.65) + \ln(50.42) \\
 & \quad \quad \quad + \ln(28.78) + \ln(11.23) \\
 & \quad \quad \quad = 15.46 \\
 & \theta = \frac{5}{15.46} = 0.3234
 \end{aligned}$$

(c) Even though we know the maximum likelihood estimate from (b), use the data in (b) to obtain the estimate by numerically maximizing the log-likelihood function using `optim` function in R. Do your answers match?

The numerical matches the theoretical.

(1d) Use the output of numerical maximization in (c) to provide approximate standard error of the maximum likelihood estimate and an approximate 95% confidence interval for θ . Are these approximations going to be good? Justify your answer

$$d) \ell''(\theta) = -\frac{n}{\theta^2}$$

$$I(\theta) = -E[\ell''(\theta)] = \frac{n}{\theta^2}$$

$$\text{Var}(\theta) = \frac{1}{I(\theta)} = \frac{\sigma^2}{n}$$

$$SE = \frac{\sigma}{\sqrt{n}} \quad 95\% \text{ CI } \hat{\theta} \pm 1.96 \cdot SE$$

$$SE = \frac{0.3234}{\sqrt{5}} = 0.1447$$

$$CI = 0.3234 \pm 1.96(0.1447) = (0.04, 0.6)$$

This is a small-sample ($n = 5$) estimate, so the normal approximation may not be very accurate.

Section 2

Code and output for 1a)

```
# Mini Project 3
x <- c(21.72, 14.65, 50.42, 28.78, 11.23)

n <- length(x)

# -----
# (a) Analytical MLE for  $\theta$ 
# -----
# Derivation:  $\hat{\theta} = n / \sum(\log(x))$ 
theta_hat <- n / sum(log(x))
cat("Analytical MLE for  $\theta$  =", round(theta_hat, 4), "\n")
```

```

>
> # -----
> # (a) Analytical MLE for  $\theta$ 
> # -----
> # Derivation:  $\theta_{\text{hat}} = n / \sum(\log(x))$ 
> theta_hat <- n / sum(log(x))
cat("Analytical MLE for  $\theta$  =", round(theta_hat, 4), "> cat("Analytical MLE
for  $\theta$  =", round(theta_hat, 4), "\n")
Analytical MLE for  $\theta$  = 0.3234

```

Code and output for 1b)

```

# -----
# (b) Verify by computing manually
# -----
sum_log_x <- sum(log(x))
theta_hat_manual <- n / sum_log_x
cat("Manual computation gives  $\theta$  =", round(theta_hat_manual, 4), "\n")

```

```

> # -----
> # (b) Verify by computing manually
> # -----
> sum_log_x <- sum(log(x))
theta_hat_manual <- n / sum_log_x
> theta_hat_manual <- n / sum_log_x
cat("Manual computation gives  $\theta$  =", round(theta_hat_manual, 4), "\n")
Manual computation gives  $\theta$  = 0.3234

```

Code and output for 1c)

```

# -----
# (c) Numerical maximization using optim()
# -----
# Log-likelihood function
loglik <- function(theta) {
  if (theta <= 0) return(-Inf)
  n * log(theta) - (theta + 1) * sum(log(x))
}

# Maximize log-likelihood
result <- optim(par = 1, fn = function(th) -loglik(th), method = "Brent", lower = 0.0001, upper = 10)
result$par

```

```

> # (c) Numerical maximization using optim()
> # -----
> # Log-likelihood function
> loglik <- function(theta) {
+   if (theta <= 0) return(-Inf)
+   n * log(theta) - (theta + 1) * sum(log(x))
+ }
>
> # Maximize log-likelihood
> result <- optim(par = 1, fn = function(th) -loglik(th), method = "Brent", lower = 0.0001, upper = 10)
> result$par
>
> result$par
[1] 0.3233874

```

Code and output for 1d)

```

# -----
# (d) Standard error and 95% CI
# -----
# Fisher Information  $I(\theta) = n / \theta^2$ 
#  $\text{Var}(\theta_{\text{hat}}) \approx \theta_{\text{hat}}^2 / n$ 
se_theta <- theta_hat / sqrt(n)
ci_lower <- theta_hat - 1.96 * se_theta
ci_upper <- theta_hat + 1.96 * se_theta

cat("Standard Error =", round(se_theta, 4), "\n")
cat("95% CI for  $\theta$ : (", round(ci_lower, 4), ",", round(ci_upper, 4), ")\n")

```

```

> # (d) Standard error and 95% CI
> # -----
> # Fisher Information  $I(\theta) = n / \theta^2$ 
> #  $\text{Var}(\theta_{\text{hat}}) \approx \theta_{\text{hat}}^2 / n$ 
> se_theta <- theta_hat / sqrt(n)
> ci_lower <- theta_hat - 1.96 * se_theta
> ci_upper <- theta_hat + 1.96 * se_theta
>
> cat("Standard Error =", round(se_theta, 4), "\n")
> cat("95% CI for  $\theta$ : (", round(ci_lower, 4), ",", round(ci_upper, 4), ")\n")
>
> cat("Standard Error =", round(se_theta, 4), "\n")
Standard Error = 0.1446
> cat("95% CI for  $\theta$ : (", round(ci_lower, 4), ",", round(ci_upper, 4), ")\n")
95% CI for  $\theta$ : ( 0.0399 , 0.6068 )

```